

SAI SAKETH BOLLA

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SUMMARY

Design-oriented Software Engineer with 3+ years of experience designing and operating **multi-cloud Kubernetes infrastructure**, **AI-driven automation pipelines**, and **high-availability telecom platforms** at scale. Proven track record architecting systems across **AWS** and **GCP** serving 30,000+ users, building observability and SRE frameworks (**Prometheus**, **Grafana**, **Cilium**), and shipping revenue-generating products.

WORK EXPERIENCE

Software Engineer (Part time)

April 2025 – Dec 2025

University of Maryland (Robust Quantum Simulation)

College Park, MD

- Architected an **AI-driven research pipeline** using **OpenAI APIs** (embeddings + completions), **Pinecone**, and semantic search to structure publication data and auto-generate reports via **python-docx/openpyxl**, powering director-level analytics with **React + FastAPI**.
- Delivered research deliverables for **IBM-QSim Conference 2025** (NYC) and **Duke Annual Conference**, automating annual reports, admin workflows, and publication tracking, reducing manual reporting effort by **80%**.

Software Engineer

Feb 2024 – Dec 2024

Prodapt Solutions

Hyderabad, India

- Designed a centralized log analysis platform ingesting **50M+ telecom logs daily** across **200+ devices**, architecting a custom indexing layer with **PostgreSQL partial indexes** and in-memory structures that reduced query latency from 5+ min to **~2s (150x improvement)**.
- Designed a **tiered storage strategy** with compression and **TTL-based retention**, cutting infrastructure costs by **45%** (~\$45K/year) while maintaining full queryability of active data.
- Shipped to **2 major telecom clients** with end-to-end ownership (**Python** collectors, **Flask** APIs, **React** dashboard), reducing MTIR from 8 hours to **1 hour** and generating **\$1.2M in attributable revenue**.

Software Engineer

May 2022 – Jan 2024

Verizon (Contract through Prodapt)

Hyderabad, India

- Designed multi-cloud **Kubernetes** infrastructure (**AWS EKS**, **GCP GKE**) serving **30,000+ users**, using **Crossplane** for declarative resource lifecycle management and **Terragrunt** for DRY IaC modules, achieving **99.5% platform uptime**.
- Engineered autoscaling with **Karpenter** (just-in-time node provisioning) and **KEDA** (event-driven pod scaling), cutting cold-start latency by **60%**; replaced kube-proxy with **Cilium (eBPF)** for kernel-level observability and **L7 policy enforcement**.
- Designed **Istio/Envoy service mesh** for mTLS and canary deployments; ran **LitmusChaos** experiments (pod kill, latency injection, node failure) to validate auto-scaling triggers and graceful degradation paths in staging.
- Governed **Major Incident Management** via **ServiceNow AIOps** with **Prometheus/Grafana** telemetry, automating anomaly detection and routing; tracked **SLO/SLI error budgets**, reducing **MTTR from 4 hours to under 30 minutes**.

PROJECTS

AI-Organized File Management | PowerShell, Windows Explorer, OpenAI (GPT-4 Mini)

- Built **FileSense**, a PowerShell tool that leverages **OpenAI GPT-4 Mini** to safely auto-sort files from Windows Explorer.
- Implemented **batch processing**, **undo system**, and **configurable alerts** to improve reliability, efficiency, and control over large folders.

Stream Command Output to Webhooks | Rust, CLI, Slack API, Google Chat API

- Built **PipeBeam**, a Rust CLI for **real-time stdout/stderr streaming** to Slack and Google Chat, enhancing script and job observability.
- Added **smart buffering**, **interactive shell mode**, and **custom alerts** to improve reliability and reduce noisy output.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Rust (Beginner)

Web & Application Development: Flask, FastAPI, React, Next.js, Node.js, REST APIs, Object-Oriented Design

Databases: PostgreSQL, MySQL, Google BigQuery, Pinecone (Vector DB)

Cloud Platforms: AWS (EKS, EC2, Lambda, S3, CloudFront), GCP (GKE, BigQuery, Cloud Composer)

Infrastructure & Orchestration: Kubernetes, Docker, Terraform, Terragrunt, Crossplane, Karpenter, KEDA

Service Mesh & Networking: Istio, Envoy, Cilium (eBPF), mTLS

Observability & SRE: Prometheus, Grafana, LitmusChaos, ServiceNow AIOps, SLO/SLI Frameworks

AI/ML Integration: OpenAI APIs (Embeddings, Completions), Semantic Search, Vector Databases

CI/CD & Testing: GitHub Actions, Jenkins, pytest, Unit/Integration Testing

Other: Linux, Git, Agile Development

Certifications: Google Cloud Associate + Google Cloud Professional DevOps

EDUCATION

University of Maryland, College Park, USA

Jan 2025 – Dec 2026

Master of Engineering in Computer Software Engineering

College Park, MD

Coursework: Software Requirements, Software Design, Virtualization & Containerization, Linux System Admin, Advanced Cloud Computing

ACHIEVEMENTS

- 4th rank in International E-school **Flipkart** grid warehouse robot making competition 2022.